

# Basant Awad Mohamed

## Backend Software Engineer

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### Professional Summary

Backend Software Engineer and final-year Computer Science graduate (Software Engineering) with hands-on experience designing REST APIs, microservices, and distributed data systems in Python, Java, and Node.js. Internship experience building production backends with Django, PostgreSQL, and Apache Kafka, plus five independent full-stack and AI-integrated projects covering cloud infrastructure, machine learning, and DevOps. Strong foundation in data structures, algorithms, system design, and software engineering best practices (SOLID, design patterns, clean architecture). Seeking an entry-level Backend Developer / Software Engineer role to apply proven problem-solving and scalable-systems skills.

### Education

#### Bachelor of Science in Computer Science — Software Engineering

Sep 2022 – Jun 2026

Al Alamein International University, Egypt

- Relevant Coursework: Data Structures, Algorithms, Distributed Systems, Database Design, Computer Networks, Operating Systems, Software Engineering, Object-Oriented Programming.

### Experience

#### Full-Stack Software Engineer Intern – Bibliotheca Alexandrina

Jul 2025 – Aug 2025

Alexandria, Egypt

- Architected a Django and PostgreSQL backend for a research publication platform, automating metadata harvesting and archiving workflows for 200+ researchers.
- Designed and built secure RESTful APIs with JWT authentication and rate-limiting, and integrated NLP-based search to improve document discovery.

#### Backend Developer Intern – Information Technology Institute (ITI)

Jul 2024 – Aug 2024

Al Alamein, Egypt

- Developed a multithreaded Java TCP/IP chat application with JDBC persistence, applying efficient data structures for real-time message routing.
- Supported concurrent client connections through thread-safe communication architecture, improving reliability under simultaneous load.

#### Student Teaching Assistant – Al Alamein International University

Sep 2025 – Jun 2026

Al Alamein, Egypt

- Supported instruction of Object-Oriented Programming, Data Structures, and Software Engineering courses for 50+ students.
- Led practical lab sessions on backend engineering principles, including SOLID, design patterns, and clean architecture.

#### Technical Course Instructor – Bianki Modern School

Jul 2025 – Aug 2025

Alexandria, Egypt

- Taught fundamentals of hardware-software interfacing and embedded logic for IoT device communication to secondary-school students.

### Projects

#### ColdBridge – Serverless Cold-Start Optimizer

Python, Rust/eBPF, Go, Kubernetes

- Built a predictive auto-scaling backend using time-series forecasting and reinforcement learning, reducing serverless cold-start latency by 45%.
- Implemented eBPF-based kernel-level monitoring for real-time, granular performance-metric collection.

#### Nexus FS – Distributed File Storage System

Node.js, Kafka, PostgreSQL, Docker

- Built event-driven microservices with Apache Kafka to handle asynchronous file processing for 10,000+ concurrent requests.
- Developed a presigned-URL service for secure file access and configured Redis caching layers using optimized search data structures.

#### AIOps Observability Platform

Laravel, Prometheus, Grafana, Machine Learning

- Built a centralized telemetry API with Prometheus integration for real-time system metrics collection.
- Developed an automated anomaly-detection pipeline that triggers server-side incident responses.

#### NovaCare – AI-Powered Assistive Rover

Node.js, MongoDB, WebSockets, BLE

- Built the backend infrastructure for an AI-powered assistive robotics system, serving multi-modal models including attention-based ASL recognition and a RAG-based conversational AI assistant.
- Engineered low-latency API endpoints for continuous BLE health-data ingestion and configured WebSockets for live dashboard telemetry.

#### FitCoach Pro – AI-Augmented Fitness Coaching Platform

- NestJS, Next.js, Flutter, PostgreSQL, Redis, BullMQ, OpenAI, Docker
- Architected a multi-tenant backend with NestJS, PostgreSQL, and Redis supporting real-time chat (Socket.io) and background job processing (BullMQ) for a fitness coaching platform serving coaches and clients.
  - Integrated AI copilot features (generative workout planning, RAG-powered advice via OpenAI and pgvector) and Stripe payment processing, containerized with Docker and managed via a Turborepo monorepo.

### Skills

- Languages:** Python, Java, SQL, JavaScript, TypeScript, Go, Rust, PHP, HTML, CSS
- Backend & APIs:** REST APIs, GraphQL, WebSockets, OpenAPI, Microservices, Django, NestJS, Node.js
- Databases:** PostgreSQL, MySQL, MongoDB, Redis
- Messaging & Streaming:** Apache Kafka, RabbitMQ
- Cloud, DevOps & Infrastructure:** Docker, Kubernetes, Terraform, CI/CD (GitHub Actions), Prometheus, Grafana, Linux, AWS
- AI & Data Science:** Machine Learning, Reinforcement Learning, Time-Series Forecasting, LLM Integration (RAG, Prompt Engineering, Embeddings), Scikit-learn, HuggingFace, Apache Airflow
- Frontend & Testing:** React, Next.js, Tailwind CSS, PyTest, JUnit, Postman, k6
- Methodologies & Core Concepts:** Agile, Scrum, Kanban, SOLID, Design Patterns, Clean Architecture, DDD, Event-Driven Architecture, Data Structures, Algorithms, OOP, System Design, Distributed Systems, Problem Solving

### Certifications

- Claude Code: Software Engineering with Generative AI Agents – Vanderbilt University, May 2026
- Introduction to DevOps – IBM, Dec 2025
- AWS Academy Graduate – Cloud Foundations – Amazon Web Services, Dec 2025
- Ethics, Technology and Engineering – 4TU.Ethics, Nov 2025 | Algorithmic Toolbox – UC San Diego, Apr 2025